1 Determinant of a Linear Operator

Exercise 1.1. Let m be a positive integer and $M \in V_{\text{alt}}^{(m)}$. Given $T \in \mathscr{L}(V)$, define $M_T \in V_{\text{alt}}^{(m)}$ by

$$M_T(v_1, \ldots, v_m) = M(Tv_1, \ldots, Tv_m)$$

for all $(v_1, \ldots, v_m) \in V^m$. Show that

(a) $M_T \in V_{\text{alt}}^{(m)}$

(b) The map $M \mapsto M_T$ is linear.

(c) If $m = n = \dim V$, then there exists a unique scalar $\alpha_T \in F$ such that $M_T = \alpha_T M$. That is,

$$M_T(v_1, \ldots, v_n) = \alpha_T M(v_1, \ldots, v_n)$$

for all $(v_1, \ldots, v_n) \in V^n$ and all $M \in V_{\text{alt}}^{(n)}$.

Definition 1.2 (Determinant of a linear operator)**.** For $T \in \mathscr{L}(V)$, the *determinant* of T, denoted by $\det T$, is defined to be the unique scalar such that $M_T = (\det T)M$ for all $M \in V_{\text{alt}}^{(\dim V)}$.

Example 1.3. Let $n = \dim V$ and let id_V be the identity operator on V. Then, for any $M \in V_{\text{alt}}^{(n)}$ and any $(v_1, \ldots, v_n) \in V^n$ we have

$$M_{\text{id}_V}(v_1, \ldots, v_n) = M(\text{id}_V v_1, \ldots, \text{id}_V v_n) = M(v_1, \ldots, v_n),$$

hence $\det(\text{id}_V) = 1$.

Exercise 1.4. Let $n = \dim V$.

(a) Show that for any $\lambda \in F$, $\det(\lambda \,\text{id}_V) = \lambda^n$.

(b) Show that, more generally, for any $\lambda \in F$ and any $T \in \mathscr{L}(V)$, $\det(\lambda T) = \lambda^n (\det T)$.

2 Determinant of a Square Matrix

Definition 2.1 (Determinant of a square matrix)**.** Let n be a positive integer and let A be an $n \times n$ matrix with entries in F. Let $T \in \mathscr{L}(M^{n\times 1}(F))$ be the operator whose matrix with respect to the standard basis of $M^{n\times 1}(F)$ equals A. The *determinant* of A, denoted by $\det A$, is defined by $\det A = \det T$.

Example 2.2. Let I be the $n \times n$ identity matrix. The corresponding operator on $M^{n\times 1}(F)$ is the identity operator, since, if e_j is the jth standard basis vector of $M^{n\times 1}(F)$, $\mathrm{id}_{M^{n\times 1}(F)} e_j = e_j$, which is the jth column of I. Thus, by Definition 2.1, $\det(I) = \det(\mathrm{id}_{M^{n\times 1}(F)}) = 1$.

Proposition 2.3. *Let n be a positive integer and $(v_1, \ldots, v_n)$ be an ordered n-tuple of column vectors. Denote by $\begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$ the matrix whose jth column is given by v_j and let $\begin{vmatrix} v_1 & \cdots & v_n \end{vmatrix}$ denote its determinant. Then the map which takes $(v_1, \ldots, v_n) \mapsto \begin{vmatrix} v_1 & \cdots & v_n \end{vmatrix}$ is an alternating n-linear form on $M^{n\times 1}(F)$.*

Proof. Let $(e_1, \ldots, e_n)$ be the standard ordered basis of $M^{n\times 1}(F)$ and let $(v_1, \ldots, v_n)$ be an ordered n-tuple of vectors in $M^{n\times 1}(F)$. By the Linear Map Lemma, there exists a unique linear map T such that $Te_j = v_j$ for each $j = 1, \ldots, n$. Thus, T is the linear operator whose matrix with respect to the standard ordered basis is $\begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$. Then $\begin{vmatrix} v_1 & \cdots & v_n \end{vmatrix} = \det(T)$ by Definition 2.1.

Let M be an alternating n-linear form on $M^{n\times 1}(F)$ such that $M(e_1, \ldots, e_n) = 1$. Then

$$\begin{aligned} \begin{vmatrix} v_1 & \cdots & v_n \end{vmatrix} &= \det(T) \\ &= \det(T) \cdot 1 \\ &= \det(T) M(e_1, \ldots, e_n) \\ &= M_T(e_1, \ldots, e_n) \quad \text{(by definition of } \det(T)) \\ &= M(Te_1, \ldots, Te_n) \quad \text{(by definition of } M_T) \\ &= M(v_1, \ldots, v_n), \end{aligned}$$

which shows that the map $(v_1, \ldots, v_n) \mapsto \begin{vmatrix} v_1 & \cdots & v_n \end{vmatrix}$ is an alternating n-linear form on $M^{n\times 1}(F)$. □

Remark 2.4. Since $\dim(M^{n\times 1}(F))^{(n)}_{\text{alt}} = 1$, a basis is given by a single nonzero alternating n-linear form, M, which is uniquely specified by $M(e_1, \ldots, e_n)$. It follows that $(v_1, \ldots, v_n) \mapsto \begin{vmatrix} v_1 & \cdots & v_n \end{vmatrix}$ is the unique n-linear form on $M^{n\times 1}(F)$ with $\begin{vmatrix} e_1 & \cdots & e_n \end{vmatrix} = 1$.

Corollary 2.5. *If an $n \times n$ matrix A has two identical columns, then* $\det(A) = 0$.

Exercise 2.6. Prove Corollary 2.5.

Proposition 2.7 (Formula for the determinant of a matrix)**.** *Let n be a positive integer and A an $n \times n$ matrix. Then*

$$\det(A) = \sum_{\sigma \in S_n} sgn(\sigma) A_{\sigma(1)1} \cdots A_{\sigma(n)n}. \tag{2.8}$$

Exercise 2.9. Prove Proposition 2.7. (*Hint*: Apply Proposition 4.4 from Worksheet 5.3.)

Example 2.10. Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$. Then $\det(A) = \sum_{\sigma \in S_2} \operatorname{sgn}(\sigma) A_{\sigma(1)1} A_{\sigma(2)2} = A_{11}A_{22} - A_{21}A_{22}$.

Exercise 2.11. Let $A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$. Use (2.8) to write out $\det(A)$.

Remark 2.12. The sum in formula 2.8 contains $n!$ terms. Because $n!$ grows rapidly as n increases, the formula in (2.8) is not a viable method to evaluate determinants even more moderately sized n. For example, 10! is over three million, and 100! is approximately 10^{158}, leading to a sum that the fastest computer cannot evaluate. We will soon see some results that lead us to faster evaluations of determinants than the direct use of the sum in (2.8).

3 Cofactor Formula for the Determinant

In the previous exercise you found that for $A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$,

$$\det A = A_{11}A_{22}A_{33} - A_{11}A_{32}A_{23} - A_{21}A_{12}A_{33} + A_{21}A_{32}A_{13} + A_{31}A_{12}A_{23} - A_{31}A_{22}A_{13}. \tag{3.1}$$

There is another way of thinking about the formula in (3.1) which is easier to remember. By collecting terms, we can write this in recursive fashion as

$$\begin{aligned} \det A &= A_{11}(A_{22}A_{33} - A_{32}A_{23}) - A_{12}(A_{21}A_{33} - A_{31}A_{23}) + A_{13}(A_{21}A_{32} - A_{31}A_{22}) \\ &= A_{11}\begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{12}\begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} + A_{13}\begin{vmatrix} A_{21} & A_{22} \\ A_{31} & A_{32}. \end{vmatrix} \end{aligned} \tag{3.2}$$

Note that $\begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix}$ is the determinant of the 2×2 sub-matrix of A obtained by deleting the first row and column: $\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$. Similarly, $\begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix}$ and $\begin{vmatrix} A_{21} & A_{22} \\ A_{31} & A_{32} \end{vmatrix}$ are the determinants of the 2×2 sub-matrices of A obtained by deleting the first row and second column and the first row and third column, respectively:

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}, \quad \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}.$$

Given a 3×3 matrix A, let $A[i|j]$ denote the 2×2 submatrix of A obtained by deleting the ith row and jth column of A and define the *ij-cofactor of A* by $C_{ij} := (-1)^{i+j}\det(A[i|j])$. Then we can write (3.2) as

$$\det A = \sum_{j=1}^{3} A_{1j}C_{1j}. \tag{3.3}$$

When using (3.3) to calculate $\det A$, one is said to be calculating the determinant of A by *cofactor expansion across the first row*.

Exercise 3.4. Calculate $\begin{vmatrix} 2 & -4 & 3 \\ 3 & 1 & 2 \\ 1 & 4 & -1 \end{vmatrix}$ by cofactor expansion across the first row.

Upon closer inspection of (3.1), one sees that there was nothing special about the first row - one can calculate $\det A$ by cofactor expansion across any row or down any column. That is, for the ith row or jth column

$$\det A = \sum_{k=1}^{3} A_{ik}C_{ik} = \sum_{k=1}^{3} A_{kj}C_{kj}.$$

This is very helpful when a matrix has many zeros.

Exercise 3.5. Calculate $\begin{vmatrix} 1 & 5 & -2 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{vmatrix}$ by cofactor expansion across the third row.

The observations above for 3×3 matrices hold for $n \times n$ matrices for any positive integer n.

Proposition 3.6. *Let n be a positive integer and let A be an $n \times n$ matrix. Then for any row i or column j*

$$\det A = \sum_{k=1}^{n} A_{ik}C_{ik} = \sum_{k=1}^{n} A_{kj}C_{kj}.$$

Proof. Let $A = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$ be an $n \times n$ matrix with columns $v_1, \ldots, v_n$. The maps $(v_1, \ldots, v_n) \mapsto \sum_{k=1}^{n} A_{ik}C_{ik}$ and $(v_1, \ldots, v_n) \mapsto \sum_{k=1}^{n} A_{kj}C_{kj}$ are both alternating n-linear forms such that $(e_1, \ldots, e_n) = 1$. Since the determinant is the unique n-linear alternating form such that $(e_1, \ldots, e_n) = 1$, we conclude that

$$\det A = \sum_{k=1}^{n} A_{ik}C_{ik} = \sum_{k=1}^{n} A_{kj}C_{kj}.$$

□

Remark 3.7. The factor $(-1)^{i+j}$ in C_{ij} determines the following checkerboard pattern of signs:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \\ + & - & + & \\ \vdots & & & \ddots \end{bmatrix}.$$

Exercise 3.8. Compute $\det A$ in the most efficient way possible, where

$$A = \begin{bmatrix} 3 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & -2 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

The matrix in Exercise 3.8 was nearly upper-triangular. In this case, the method of that exercise can be adapted to prove the following theorem.

Proposition 3.9. *If A is a triangular matrix with $\lambda_1, \ldots, \lambda_n$ on the diagonal, then* $\det A = \lambda_1 \cdots \lambda_n$.

Exercise 3.10. Prove Proposition 3.9 either by adopting the method of Exercise 3.8 or directly from the formula in (2.8).